

Moments and Singular Perturbation

Yoonjae Hwang

September 18, 2025

1 Problem Setup

SYSTEM:

$$\dot{x} = A_{11}x + A_{12}z + B_1u \quad (1)$$

$$\varepsilon \dot{z} = A_{21}x + A_{22}z + B_2u \quad (2)$$

$$y = C_1x \quad (3)$$

No z term as otherwise we would have feedforward in the reduced model.

In state space form:

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} B_1 \\ \frac{1}{\varepsilon}B_2 \end{bmatrix} u \quad (4)$$

SIGNAL:

$$\dot{w}_1 = S_{11}w_1 + S_{12}w_2 \quad (5)$$

$$\varepsilon \dot{w}_2 = S_{21}w_1 + S_{22}w_2 \quad (6)$$

REDUCED MODEL SIGNAL:

$$S_{22}w_2 + S_{21}w_1 = 0 \quad (7)$$

$$w_2 = -S_{22}^{-1}S_{21}w_1 \quad (8)$$

$$\dot{w}_1 = (S_{11} - S_{12}S_{22}^{-1}S_{21})w_1 \quad (9)$$

$$u = (L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \quad (10)$$

REDUCED MODEL OF THE SYSTEM:

$$A_{22}z + A_{21}x + B_2u = 0 \quad (11)$$

$$z = -A_{22}^{-1}(A_{21}x + B_2u) \quad (12)$$

$$\dot{x} = (A_{11} - A_{12}A_{22}^{-1}A_{21})x + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \quad (13)$$

FORWARD MOMENTS

$$M_g := [C_1\Pi_{x1} \quad C_1\Pi_{x2}] \quad (14)$$

FORWARD SYLVESTER EQUATION FOR THE FULL SYSTEM:

$$\begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} + \begin{bmatrix} B_1L_{g1} & B_1L_{g2} \\ \frac{1}{\varepsilon}B_2L_{g1} & \frac{1}{\varepsilon}B_2L_{g2} \end{bmatrix} = \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} \\ \frac{1}{\varepsilon}S_{21} & \frac{1}{\varepsilon}S_{22} \end{bmatrix} \quad (15)$$

FORWARD SYLVESTER EQUATION FOR THE REDUCED MODEL

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})\tilde{\Pi} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) = \tilde{\Pi}(S_{11} - S_{12}S_{22}^{-1}S_{21}) \quad (16)$$

Can we show that as $\varepsilon \rightarrow 0$, the forward sylvester equation for the full system tends towards the forward sylvester equation for the reduced model?

2 Proof (I guess?)

Expanding out (14) we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \Pi_{x1}S_{11} + \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \quad (17a)$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12} + \frac{1}{\varepsilon}\Pi_{x2}S_{22}, \quad (17b)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x1} + A_{22}\Pi_{z1}) + \frac{1}{\varepsilon}B_2L_{g1} = \Pi_{z1}S_{11} + \frac{1}{\varepsilon}\Pi_{z2}S_{21}, \quad (17c)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x2} + A_{22}\Pi_{z2}) + \frac{1}{\varepsilon}B_2L_{g2} = \Pi_{z1}S_{12} + \frac{1}{\varepsilon}\Pi_{z2}S_{22}. \quad (17d)$$

Multiplying (16c) and (16d) by ε we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \Pi_{x1}S_{11} + \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \quad (18a)$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12} + \frac{1}{\varepsilon}\Pi_{x2}S_{22}, \quad (18b)$$

$$A_{21}\Pi_{x1} + A_{22}\Pi_{z1} + B_2L_{g1} = \varepsilon\Pi_{z1}S_{11} + \Pi_{z2}S_{21}, \quad (18c)$$

$$A_{21}\Pi_{x2} + A_{22}\Pi_{z2} + B_2L_{g2} = \varepsilon\Pi_{z1}S_{12} + \Pi_{z2}S_{22}. \quad (18d)$$

We now consider the ε series expansion of Π_{x1} , Π_{x2} , Π_{z1} , Π_{z2} . We can see that by multiplying (17a) and (17b) by ε , and setting ε to 0 we have that the zeroth order term for Π_{x2} is 0.

We then write:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x1}^{(0)}S_{11} + \Pi_{x2}^{(1)}S_{21} + O(\varepsilon), \quad (19a)$$

$$A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} = \Pi_{x1}^{(0)}S_{12} + \Pi_{x2}^{(1)}S_{22} + O(\varepsilon), \quad (19b)$$

$$A_{21}\Pi_{x1}^{(0)} + A_{22}\Pi_{z1}^{(0)} + B_2L_{g1} = \Pi_{z2}^{(0)}S_{21} + O(\varepsilon), \quad (19c)$$

$$A_{22}\Pi_{z2}^{(0)} + B_2L_{g2} = \Pi_{z2}^{(0)}S_{22} + O(\varepsilon) \quad (19d)$$

Each of the 4 equations in (18) are essentially polynomials in epsilon.

After equating coefficients, we can write $\Pi_{x2}^{(1)}$ as such (Assuming S_{22} is invertible):

$$\Pi_{x2}^{(1)} = \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} - \Pi_{x1}^{(0)}S_{12} \right) S_{22}^{-1} \quad (20)$$

Substituting (19) into (18a) we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x1}^{(0)}S_{11} + \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} - \Pi_{x1}^{(0)}S_{12} \right) S_{22}^{-1}S_{21} + O(\varepsilon) \quad (21)$$

Rearranging we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} \right) S_{22}^{-1}S_{21} + O(\varepsilon) \quad (22)$$

Similarly, from (18c) we can substitute in an expression for $\Pi_{z1}^{(0)}$:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}A_{22}^{-1} \left(\Pi_{z2}^{(0)}S_{21} - A_{21}\Pi_{x1}^{(0)} - B_2L_{g1} \right) + B_1L_{g1} = \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} \right) S_{22}^{-1}S_{21} + O(\varepsilon) \quad (23)$$

Rearranging we have:

$$\begin{aligned} & (A_{11} - A_{12}A_{22}^{-1}A_{21})\Pi_{x1}^{(0)} + (B_1 - A_{12}A_{22}^{-1}B_2)L_{g1} \\ & + \left(A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - \left(A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} \right) S_{22}^{-1} \right) S_{21} \\ & = \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + O(\varepsilon) \end{aligned} \quad (24)$$

Simplifying the middle (...) S_{21} term using (18d):

$$\left(A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - A_{12}\Pi_{z2}^{(0)}S_{22}^{-1} - B_1L_{g2}S_{22}^{-1} \right) S_{21} \quad (25)$$

$$= \left(A_{12}A_{22}^{-1} \left(\Pi_{z2}^{(0)}S_{22} - A_{22}\Pi_{z2}^{(0)} \right) S_{22} - B_1L_{g2}S_{22}^{-1} \right) S_{21} \quad (26)$$

$$= \left(A_{12}A_{22}^{-1}B_2L_{g2}S_{22}^{-1} - B_1L_{g2}S_{22}^{-1} \right) S_{21} \quad (27)$$

$$= - \left(B_1 - A_{12}A_{22}^{-1}B_2 \right) L_{g2}S_{22}^{-1}S_{21} \quad (28)$$

Substituting (27) back into (23) and factorising we have:

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})\Pi_{x1}^{(0)} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) \quad (29)$$

$$= \Pi_{x1}^{(0)}(S_{11} - S_{12}S_{22}^{-1}S_{21}) + O(\varepsilon) \quad (30)$$

which is of $O(\varepsilon)$ error from (15). Thus as $\varepsilon \rightarrow 0$, (29) approaches (15), and the matrix $\Pi_{x1} \rightarrow \Pi_{x1}^{(0)} = \tilde{\Pi}$.

3 Next Steps

- Expression for Π_{x2} (Is it needed?)
- Higher order epsilon expansion?
- Assumptions, and rigour?

4 S_{11} and S_{22} are 0

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \quad (31a)$$

$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12}, \quad (31b)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x1} + A_{22}\Pi_{z1}) + \frac{1}{\varepsilon}B_2L_{g1} = \frac{1}{\varepsilon}\Pi_{z2}S_{21}, \quad (31c)$$

$$\frac{1}{\varepsilon}(A_{21}\Pi_{x2} + A_{22}\Pi_{z2}) + \frac{1}{\varepsilon}B_2L_{g2} = \Pi_{z1}S_{12}. \quad (31d)$$

Similar approach to before, we'll have $\Pi_{x2}^{(0)} = 0$.

Considering zeroth order terms:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x2}^{(1)}S_{21}, \quad (32a)$$

$$A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} = \Pi_{x1}^{(0)}S_{12}, \quad (32b)$$

$$A_{21}\Pi_{x1}^{(0)} + A_{22}\Pi_{z1}^{(0)} + B_2L_{g1} = \Pi_{z2}^{(0)}S_{21}, \quad (32c)$$

$$A_{22}\Pi_{z2}^{(0)} + B_2L_{g2} = 0 \quad (32d)$$

Assuming A_{22} is invertible:

$$\Pi_{z2}^{(0)} = -A_{22}^{-1}B_2L_{g2} \quad (33)$$

Assuming S_{12} is invertible, (33) $\rightarrow$ (32b):

$$\Pi_{x1}^{(0)} = (-A_{12}A_{22}^{-1}B_2L_{g2} + B_1L_{g1})S_{12}^{-1} \quad (34)$$

And $\Pi_{z1}^{(0)}$ can be extracted from (32c). Simulation results doesnt seem to agree. Might be implementation issue.

5 Thoughts

- Canonical form - is there a coordinate change that we can exploit?
- Making a family of approximations? what to do when A_{22} is not Hurwitz?
- Realisation?
- Controllability and observability conditions
- Is there a limit for the difference? $\Pi w_g - x$

6 Singular Systems

One thing to note that you could represent singular perturbation systems as a singular system. Scarlotti wrote a paper on singular systems and moment matching, but something different is that for us, our signal is also a singular perturbation system. Another thing to note, when you consider the scenario where S_{11} and S_{22} are 0, the signal is a pure integrator, but also when you set $\varepsilon = 0$, you have $S_{21}w_1 = 0$ which is ambiguous. A reduced model for the signal is not well defined. So a better approach might be to consider finite ε and then take the limit as $\varepsilon \rightarrow 0$.

6.1 New Definition of Moments

I think a better definition for the matrix Π is to consider it as a linear transformation of the signal w , such that the difference between Πw and x is governed by the open loop system dynamics. Then, the Sylvester equation becomes a condition that Π must satisfy for it to be a valid linear transformation. This allows us to move away from the frequency domain definition of moments which relies on the system having a steady state response. Even for unstable systems, we can still compute a solution for the Sylvester equation when there is no steady state response.

$$\frac{d}{dt}(x - \Pi w) = A(x - \Pi w) \quad (35)$$

This way a nonlinear extension should be more straightforward, as Π will probably be some nonlinear operation on w .

7 Singular Systems and Singular Perturbation

A singular system is defined as:

$$E\dot{x} = Ax + Bu \quad (36)$$